

Challenge (Habanero)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x$, what is $(f + g)(x)$?
(A) $x^2 + 2x$
(B) $x^2 - 2x$
(C) $2x^2$
(D) $x^2 + x$
(E) $2x^2 + x$
- Q2.** What is the domain of $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-1}$ when combined as $(f \cdot g)(x)$?
(A) $x \neq 0$
(B) $x \neq 1$
(C) $x \neq 0, x \neq 1$
(D) $x > 0$
(E) $x < 0$
- Q3.** If $f(x) = x + 1$ and $g(x) = x - 1$, what is $(f - g)(x)$?
(A) 2
(B) $2x$
(C) $x + 2$
(D) $x - 2$
(E) $2x + 1$
- Q4.** Given $f(x) = x^2$ and $g(x) = x + 1$, simplify $(f \cdot g)(x)$ to...
(A) $x^3 + x^2$
(B) $x^3 + x$
(C) $x^2 + x$
(D) $x^2 + 1$
(E) $x^3 - x^2$
- Q5.** What is the result of $(f + g)(x)$ if $f(x) = 2x$ and $g(x) = -x$?
(A) x
(B) $2x$
(C) $-x$
(D) $3x$
(E) x^2
- Q6.** If $f(x) = \frac{1}{x}$ and $g(x) = x$, what is $(f \cdot g)(x)$?
(A) 1
(B) x
(C) $\frac{1}{x}$
(D) x^2
(E) $\frac{x}{x}$
- Q7.** Determine the domain of $(f - g)(x)$ where $f(x) = x$ and $g(x) = \frac{1}{x}$.
(A) $x \neq 0$
(B) $x > 0$
(C) $x < 0$
(D) $x = 0$
(E) $x \neq 1$
- Q8.** Given $f(x) = x^2 - 1$ and $g(x) = x + 1$, simplify $(f + g)(x)$ to...
(A) $x^2 + x$
(B) $x^2 + x + 1$
(C) $x^2 - x - 1$
(D) $x + 1$
(E) $x^2 + 2x + 1$

Open Ended Questions (FRQ)

- Q9.** Given two functions $f(x) = x^2$ and $g(x) = 2x - 1$, determine the expression and domain for $(f \cdot g)(x)$. Show your work.
- Q10.** A game developer is creating a scoring system where $f(x) = x + 10$ represents points from completing a level and $g(x) = 2x$ represents points from collecting coins. Determine the combined score function $(f + g)(x)$ and its domain if x cannot be negative.

Chef's Tips

- Provide clear instructions for each question.
- Ensure students understand the context of each problem.
- Allow the use of calculators only for specific questions where it's explicitly stated.